

Packet Analysis with Wireshark Report

Task Objective

The objective of this task was to capture and analyze network traffic using Wireshark. Different protocols including HTTP, FTP, and DNS were examined. FTP credentials were extracted from unencrypted traffic, and a TCP SYN flood attack was simulated using hping3 to observe its behavior on the network.

Tools Used

Tool	Purpose
Kali Linux	Attack machine and packet capture
Metasploitable2	Vulnerable target machine
Wireshark	Packet capture and protocol analysis
hping3	SYN flood simulation
FTP Server (Metasploitable2)	Credential analysis
Browser/Terminal	HTTP and FTP traffic generation

Lab Environment

Component	Details
Attacker Machine	Kali Linux
Target Machine	Metasploitable2
Kali IP Address	192.168.180.132
Target IP Address	192.168.180.131
Network	VMware NAT (or Host-only, whichever you used)

Part A – HTTP Traffic Analysis

Objective

Capture HTTP communication between Kali Linux and the target system.

Procedure

1. Started packet capture on interface **eth0**.
2. Applied the display filter: http
3. Generated HTTP traffic by opening the target web server.
4. Observed HTTP requests and responses.

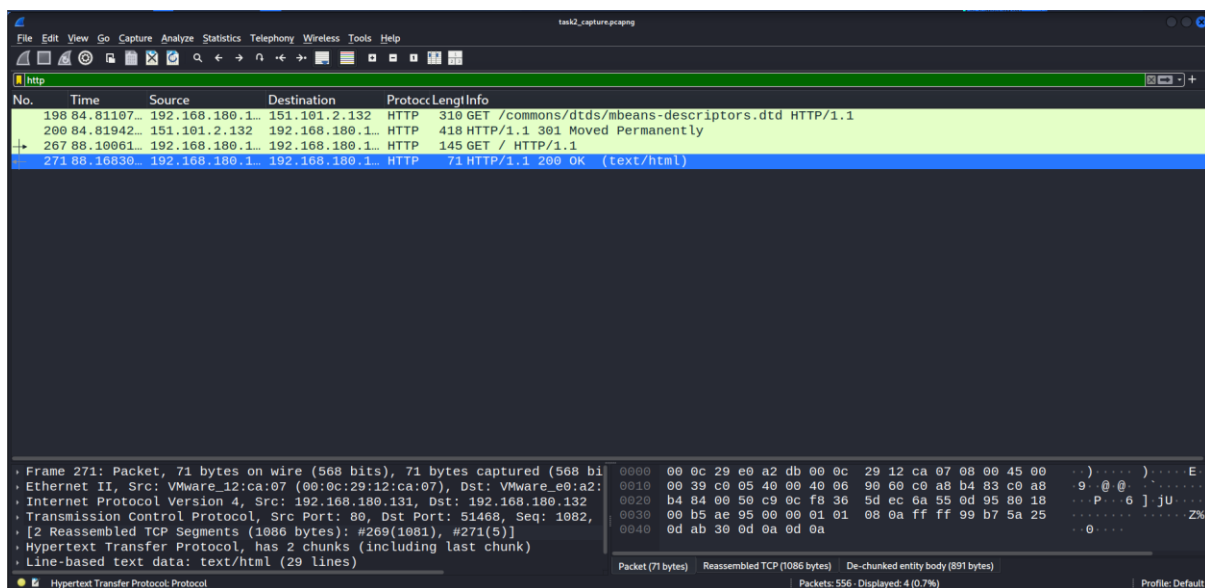
Observation

The capture showed HTTP GET requests and corresponding HTTP responses transmitted in plaintext.

Wireshark Filter Used

http

Screenshot



Analysis

HTTP is an unencrypted application layer protocol. Since data is transmitted in plaintext, URLs, headers, and other information can easily be viewed by anyone capturing the network traffic.

Part B – FTP Traffic Analysis

Objective

Capture FTP communication and observe FTP commands.

Procedure

1. Started packet capture.
2. Connected to the FTP server running on Metasploitable2.
3. Logged in using valid FTP credentials.
4. Applied the display filter: ftp

Observation

The Wireshark capture displayed FTP commands including:

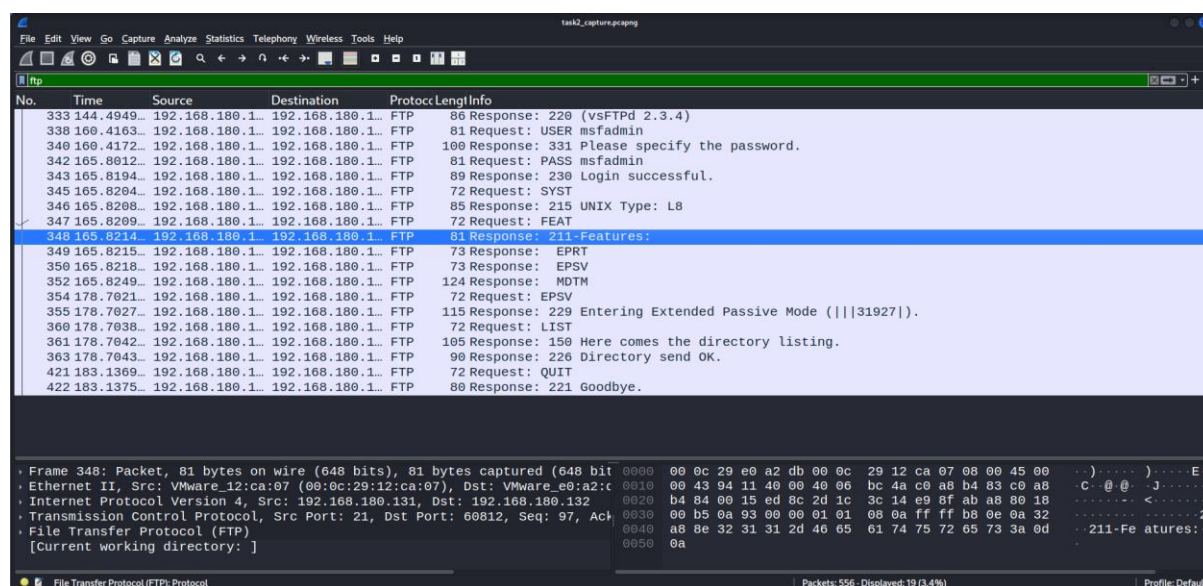
- USER
- PASS
- LIST
- RETR

All FTP commands were transmitted without encryption.

Wireshark Filter

ftp

Screenshot



Part C – Extracting FTP Credentials

Objective

Demonstrate how credentials can be captured from unencrypted FTP traffic.

Procedure

Applied the following display filter:

ftp.request.command=="USER" || ftp.request.command=="PASS"

or simply searched for PASS

Observation

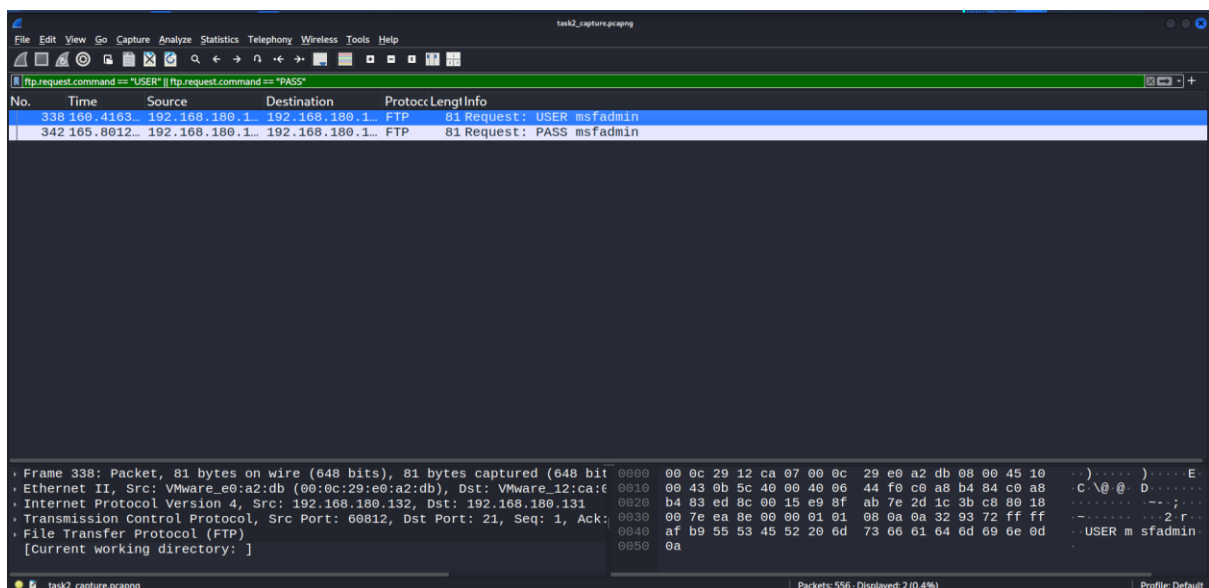
The username and password were visible in plaintext inside FTP packets.

Example:

USER msfadmin

PASS msfadmin

Screenshot



Analysis

Since FTP does not encrypt authentication data, attackers monitoring network traffic can easily capture usernames and passwords. This highlights why secure alternatives such as SFTP or FTPS are recommended.

Part D – DNS Traffic Analysis

Objective

Analyze DNS queries generated during communication.

Procedure

Applied the display filter

dns

Generated DNS requests by accessing a website or performing DNS resolution.

Observation

Wireshark displayed:

- Standard DNS Query
- Standard DNS Response

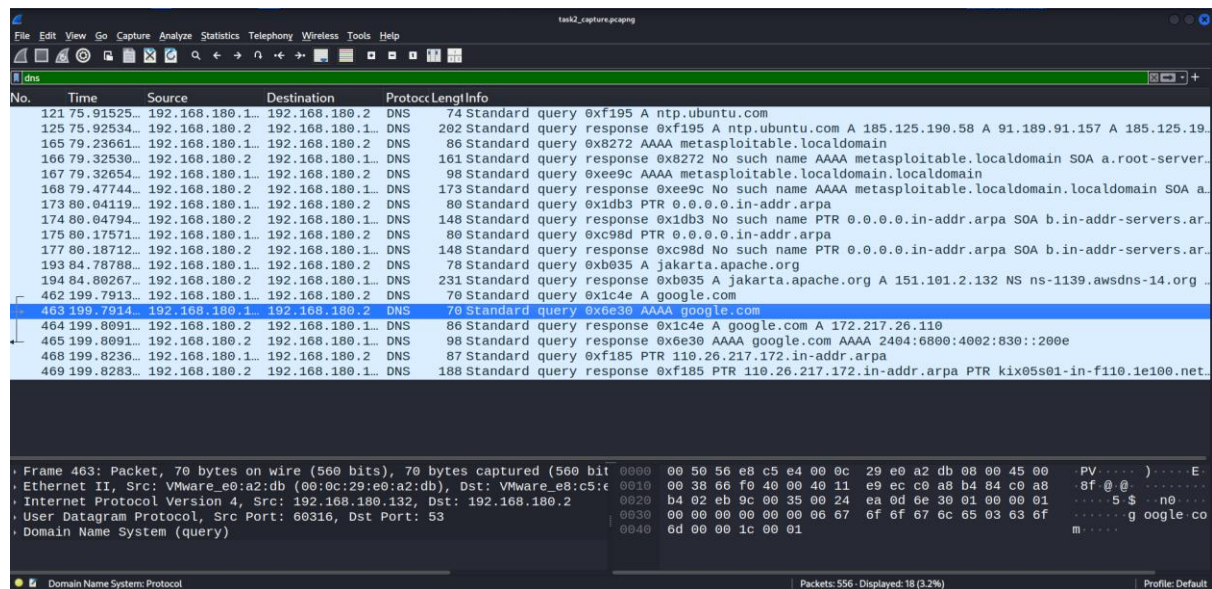
The packets contained:

- Queried domain name
- Record type (A)
- Returned IP address

Wireshark Filter

dns

Screenshot



Analysis

DNS resolves domain names into IP addresses before communication begins. Wireshark allows inspection of DNS queries, response codes, and resolved addresses.

Part E – SYN Flood Attack Analysis

Objective

Observe characteristics of a TCP SYN flood attack.

Tool Used

hping3

Command Used

```
sudo hping3 -S -p 80 --flood 192.168.180.131
```

Wireshark Display Filter

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

Procedure

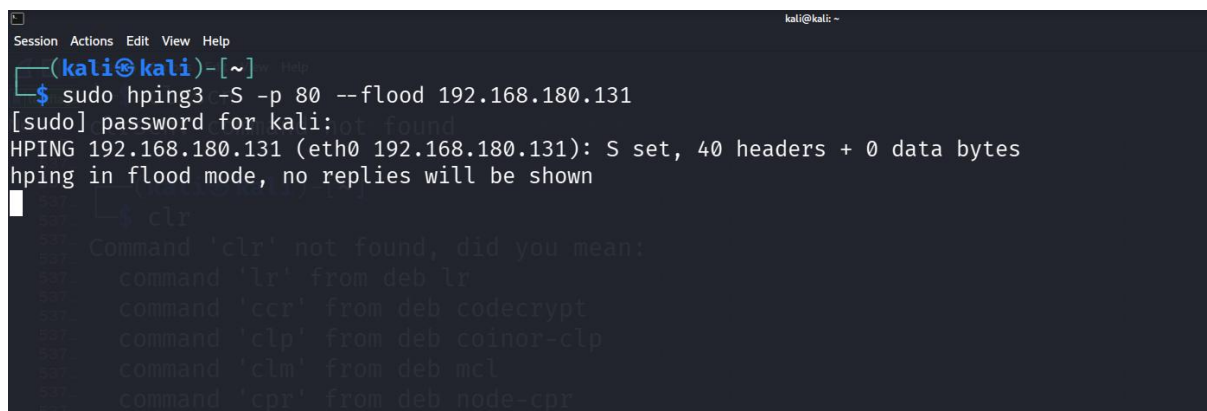
1. Started packet capture on Kali Linux.
2. Executed the hping3 SYN flood command.
3. Allowed the attack to run for approximately 5–10 seconds.
4. Stopped packet capture.
5. Applied the SYN packet filter.

Observation

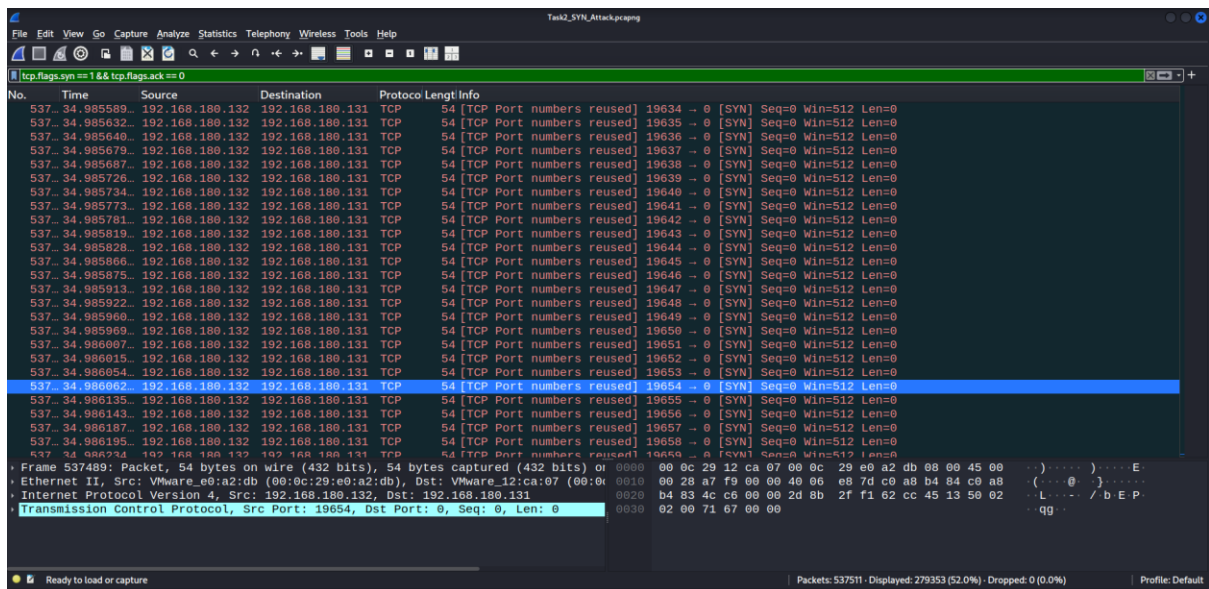
Wireshark displayed a continuous stream of TCP SYN packets sent from the Kali Linux machine to the Metasploitable2 target.

No application-layer communication occurred because only SYN packets were transmitted repeatedly.

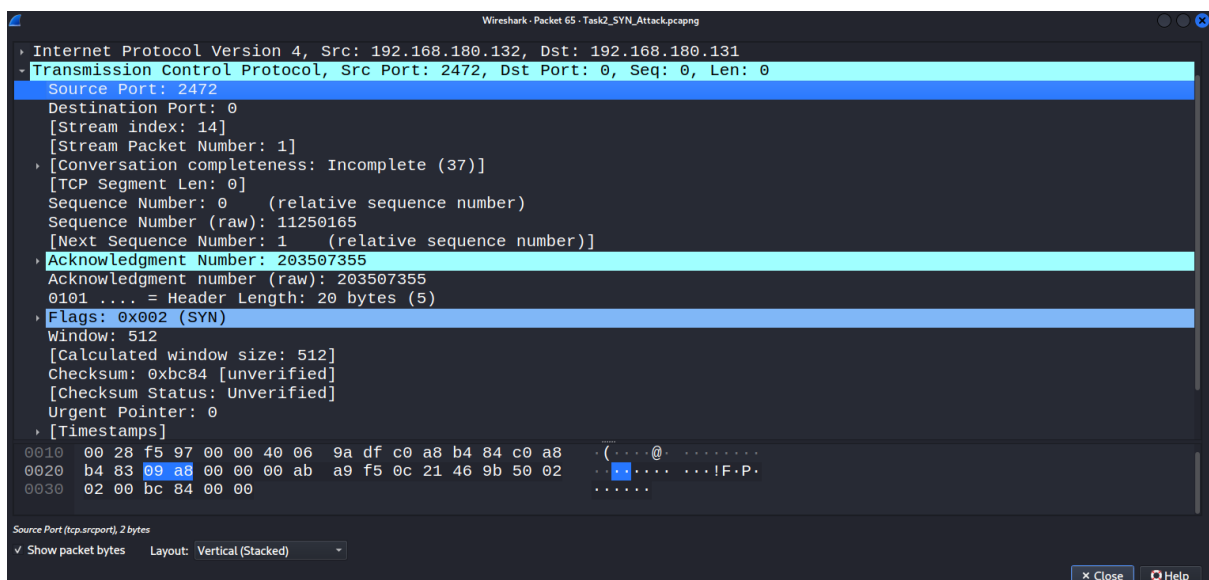
Screenshot

A screenshot of a terminal window on a Kali Linux system. The terminal shows the command 'sudo hping3 -S -p 80 --flood 192.168.180.131' being entered and executed. The output shows 'HPING 192.168.180.131 (eth0 192.168.180.131): S set, 40 headers + 0 data bytes' and 'hping in flood mode, no replies will be shown'. The user then enters 'ctrl' which results in a message: 'Command 'clr' not found, did you mean: command 'lr' from deb lr, command 'ccr' from deb codecrypt, command 'clp' from deb coinor-clp, command 'clm' from deb mcl, command 'cpr' from deb node-cpr'.

(hping3 terminal running)



(Wireshark showing many SYN packets)



(Expanded TCP header showing SYN flag set)

Analysis

A SYN flood attack repeatedly sends TCP SYN packets without completing the TCP three-way handshake. This forces the target system to allocate resources for half-open connections, potentially exhausting available resources and causing a denial-of-service condition. Wireshark clearly illustrated the large volume of SYN packets generated during the simulation.

Conclusion

Wireshark successfully captured and analyzed HTTP, FTP, DNS, and TCP traffic in a controlled virtual lab environment. The analysis demonstrated the insecurity of plaintext protocols such

as FTP and HTTP, allowing credentials and application data to be viewed directly from captured packets. The SYN flood simulation highlighted how network-based denial-of-service attacks appear in packet captures and emphasized the importance of traffic analysis for detecting malicious network behavior.